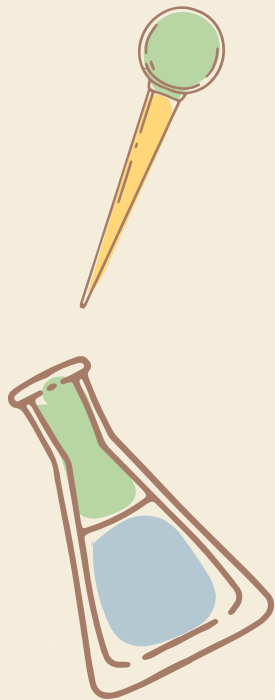
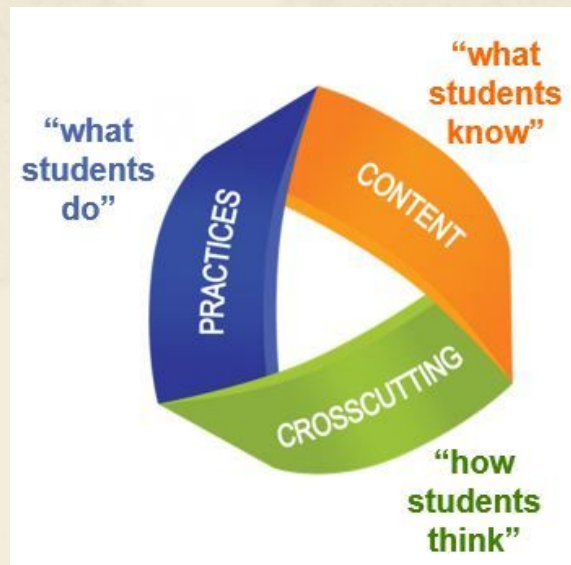


$$\pi = 3,141592$$



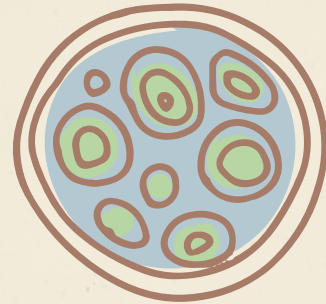
NGSS Overview: Supporting Instruction and Student Learning



October 2022

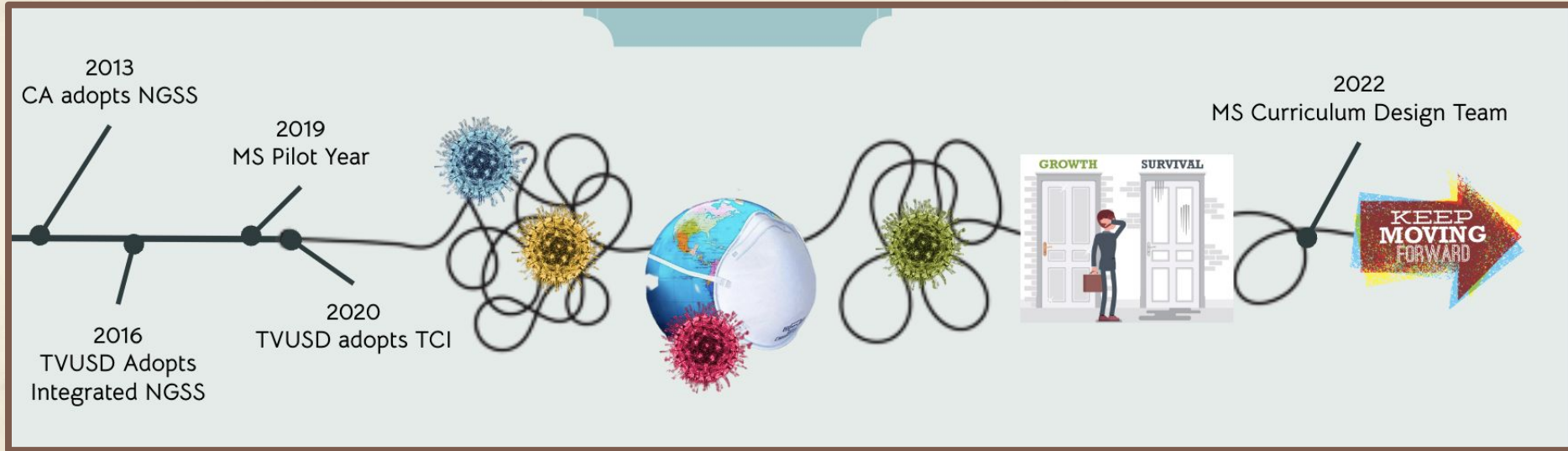
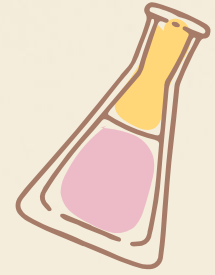


Importance of NGSS Standards



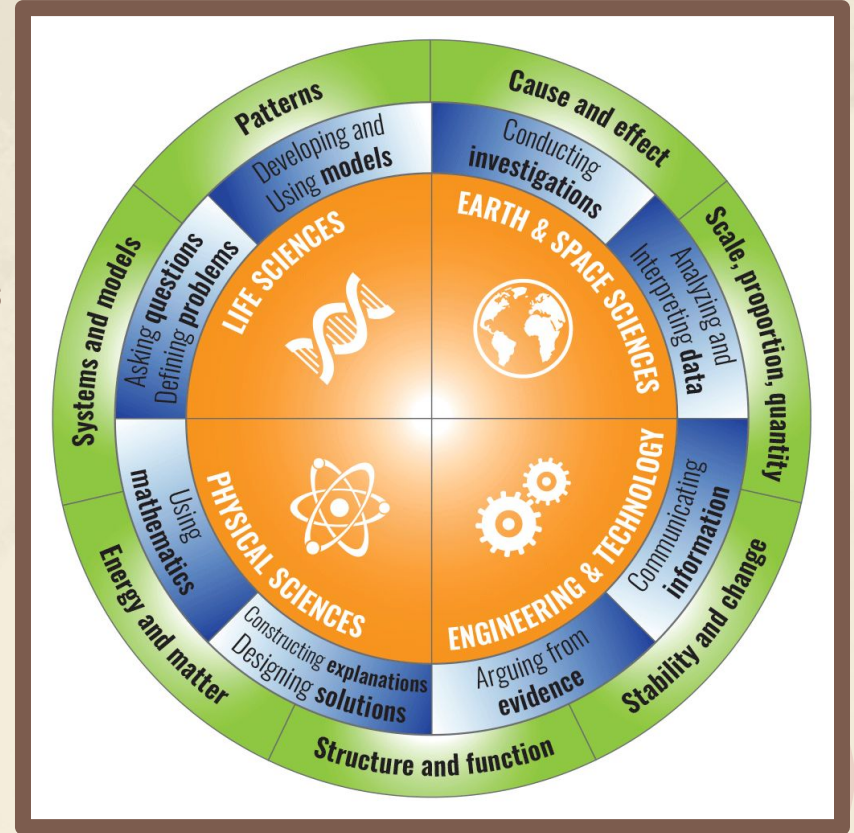


Our Journey So Far



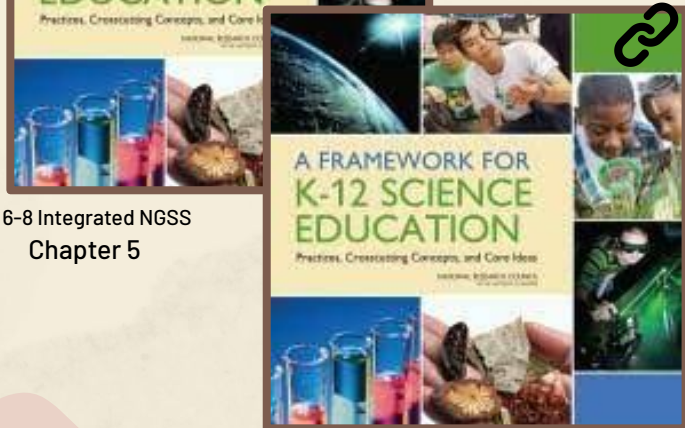
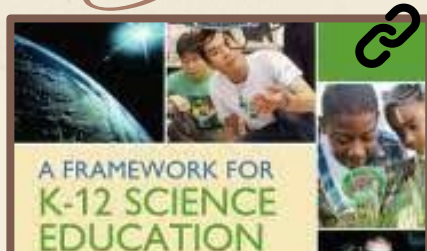
Today's Objectives

1. Understand NGSS Framework/Standards
2. Understand Instructional Shifts in Science Classrooms
3. Action Steps for Supporting Changes with instruction and learning in science courses
4. Equity





NGSS Framework and Standards



6-8 Integrated NGSS
Chapter 5

9-12 3 Course Model Integrated NGSS
Chapter 7



"Unless otherwise specified, 'descriptions' referenced in the evidence statements could include but are not limited to written, oral, pictorial, and kinesthetic descriptions."

HS-LS1-1

Students who demonstrate understanding can:

HS-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. *[Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.]*

The performance expectation above was developed using the following elements from *A Framework for K-12 Science Education*:

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9–12 builds on K–8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS1.A: Structure and Function - Systems of specialized cells within organisms help them perform the essential functions of life. - All cells contain genetic information in the form of DNA molecules. Genes are regions in the DNA that contain the instructions that code for the formation of proteins, which carry out most of the work of cells. <i>(Note: This Disciplinary Core Idea is also addressed by HS-LS3-1.)</i>	Structure and Function Investigating or designing new systems or structures requires a detailed examination of the properties of different materials, the structures of different components, and connections of components to reveal its function and/or solve a problem.



Scientific and
Engineering
Practices (SEP)



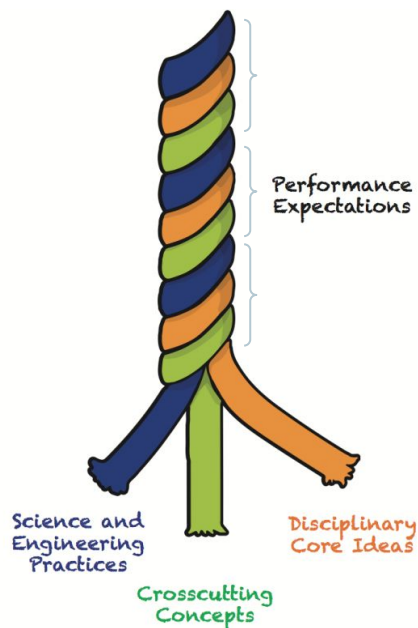
Disciplinary Core Ideas



Crosscutting
Concepts

Next Generation Science Standards

Three-Dimensional Learning



Science and Engineering Practices

- ★ 1. Asking questions (for science) and defining problems (for engineering)
- ★ 2. Developing and using models
- ★ 3. Planning and carrying out investigations
- ★ 4. Analyzing and interpreting data
- ★ 5. Using mathematics and computational thinking
- ★ 6. Constructing explanations (for science) and designing solutions (for engineering)
- ★ 7. Engaging in argument from evidence
- ★ 8. Obtaining, evaluating, and communicating information

★ Standards directly related to math and ELA

PRACTICES

CORE IDEAS

CROSSCUTTING

Disciplinary Core Ideas

- Life Sciences
- Physical Sciences
- Earth Science
- Engineering, Technology, and Application of Science

Crosscutting Concepts

- ★ 1. Patterns.
- ★ 2. Cause and effect: Mechanism and Explanation
- ★ 3. Scale, proportion, and quantity
- ★ 4. Systems and system models
- 5. Energy and matter: Flows, Cycles, and Conservation
- 6. Structure and function
- ★ 7. Stability and change



More on Science and Engineering Practices (SEP)

Science and Engineering Practices

The multiple ways of knowing and doing that scientists and engineers use to study the natural world and design world.

1. Asking questions and defining problems
2. Developing and using models
3. Planning and carrying out investigations and designing solutions
4. Analyzing and interpreting data
5. Using mathematics and computational thinking
6. Constructing explanations and designing solutions
7. Engaging in argument from evidence
8. Obtaining, evaluating, and communicating information



The practices work together – they are not separated!

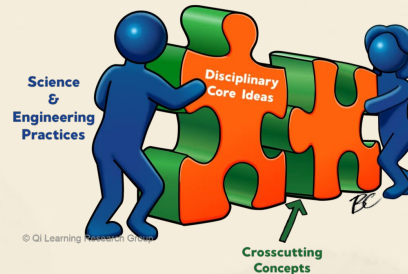
18

SEPs Change Science Education

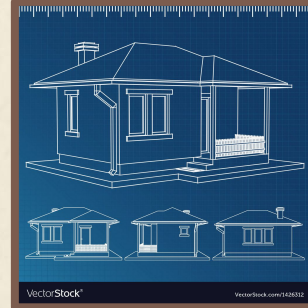
From the science classroom as environments where students **learn about science ideas**

TO

Places where students **explore, examine, and use science ideas to figure out how and why phenomena occur.**



More on Crosscutting Concepts



Crosscutting Concepts

1. Cause and Effect
2. Structure and Function
3. Systems and System Models
4. Scale, Proportion, and Quantity
5. Stability and Change
6. Energy and Matter
7. Patterns

Tch

These concepts help provide students with an organizational framework for connecting knowledge from the various disciplines into a coherent and scientifically based view of the world.

Framework, p. 4-1



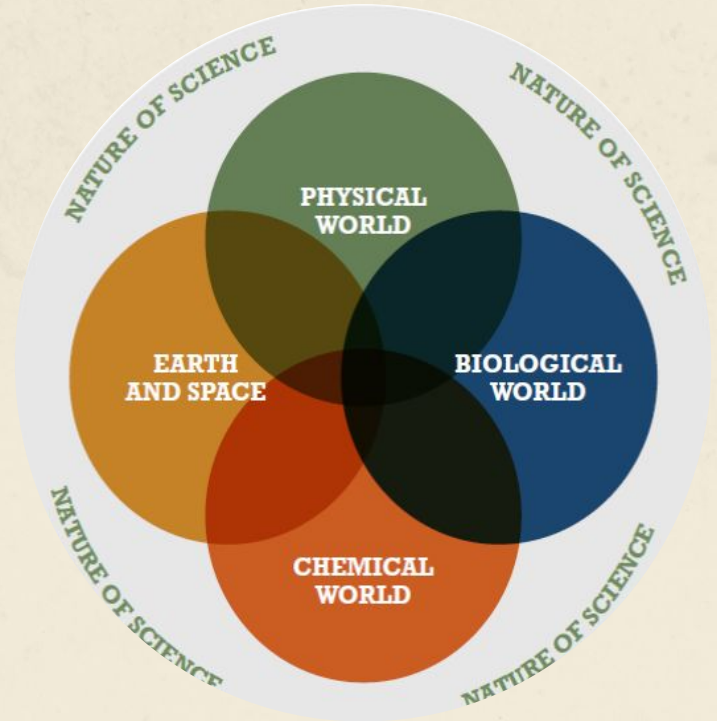
More on Disciplinary Core Ideas (DCI)



Life Science	Earth & Space Science	Physical Science	Engineering & Technology
LS1: From Molecules to Organisms: Structures and Processes LS1.A: Structure and Function LS1.B: Growth and Development of Organisms LS1.C: Organization for Matter and Energy Flow in Organisms LS1.D: Information Processing LS2: Ecosystems: Interactions, Energy, and Dynamics LS2.A: Interdependent Relationships in Ecosystems LS2.B: Cycles of Matter and Energy Transfer in Ecosystems LS2.C: Ecosystem Dynamics, Functioning, and Resilience LS2.D: Social Interactions and Group Behavior LS3: Heredity: Inheritance and Variation of Traits LS3.A: Inheritance of Traits LS3.B: Variation of Traits LS4: Biological Evolution: Unity and Diversity LS4.A: Evidence of Common Ancestry and Diversity LS4.B: Natural Selection LS4.C: Adaptation LS4.D: Biodiversity and Humans	ESS1: Earth's Place in the Universe ESS1.A: The Universe and Its Stars ESS1.B: Earth and the Solar System ESS1.C: The History of Planet Earth ESS2: Earth's Systems ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS2.C: The Roles of Water in Earth's Surface Processes ESS2.D: Weather and Climate ESS2.E: Biogeology ESS3: Earth and Human Activity ESS3.A: Natural Resources ESS3.B: Natural Hazards ESS3.C: Human Impacts on Earth Systems ESS3.D: Global Climate Change	PS1: Matter and Its Interactions PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions PS1.C: Nuclear Processes PS2: Motion and Stability: Forces and Interactions PS2.A: Forces and Motion PS2.B: Types of Interactions PS2.C: Stability and Instability in Physical Systems PS3: Energy PS3.A: Definitions of Energy PS3.B: Conservation of Energy and Energy Transfer PS3.C: Relationship Between Energy and Forces PS3.D: Energy in Chemical Processes and Everyday Life PS4: Waves and Their Applications in Technologies for Information Transfer PS4.A: Wave Properties PS4.B: Electromagnetic Radiation PS4.C: Information Technologies and Instrumentation	ETS1: Engineering Design ETS1.A: Defining and Delimiting an Engineering Problem ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution ETS2: Links Among Engineering, Technology, Science, and Society ETS2.A: Interdependence of Science, Engineering, and Technology ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World <i>Note: In NGSS, the core ideas for Engineering, Technology, and the Application of Science are integrated with the Life Science, Earth & Space Science, and Physical Science core ideas</i>

What is INTEGRATED NGSS?

2 or more science strands tied to a concept or phenomena



Anatomy of an NGSS PE:

Science
Strand

PE #-Sub topic#

Grade

PE: 3-Dimensions of what students should be able to DO at the end of the UNIT

Where to FOCUS instruction

Students who demonstrate understanding can:

MS-PS1-1.

Develop models to describe the atomic composition of simple molecules and extended structures.

[Clarification]

Statement: Emphasis is on developing models of molecules that vary in complexity. Examples of simple molecules could include ammonia and methanol. Examples of extended structures could include sodium chloride or diamonds. Examples of molecular-level models could include drawings, 3D ball and stick structures, or computer representations showing different molecules with different types of atoms. [Assessment Boundary: Assessment does not include valence electrons and bonding energy, discussing the ionic nature of subunits of complex structures, or a complete description of all individual atoms in a complex molecule or extended structure is not required.]

Where to STOP instruction

Skill: SWBAT DO...

the expectation above was developed

Content: SWBAT Know...

document A Framework

Lense: SWBAT Identify/ View through the lense

Science and Engineering Practices

Developing and Using Models

Modeling in 6–8 builds on K–5 and progresses to developing, using and revising models to describe, test, and predict more abstract phenomena and design systems.

- Develop a model to predict and/or describe phenomena.

Disciplinary Core Ideas

PS1.A: Structure and Properties of Matter

- Substances are made from different types of atoms, which combine with one another in various ways. Atoms form molecules that range in size from two to thousands of atoms.
- Solids may be formed from molecules, or they may be extended structures with repeating subunits (e.g., crystals).

Crosscutting Concepts

Scale, Proportion, and Quantity

- Time, space, and energy phenomena can be observed at various scales using models to study systems that are too large or too small.

Connections to other DCIs in this grade-band:

MS.ESS2.C

Articulation of DCIs across grade-bands:

5.PS1.A : HS.PS1.A : HS.ESS1.A

Connections to other DCIs in this grade level

Vertical Articulation of DCIs across grade levels

Common Core State Standards Connections:

ELA/Literacy -

RST.6-8.7

Integrate quantitative or technical information expressed in words in a text with a version of that information expressed visually (e.g., in a flowchart, diagram, model, graph, or table). (MS-PS1-1)

Mathematics -

MP.2

Reason abstractly and quantitatively. (MS-PS1-1)

MP.4

Model with mathematics. (MS-PS1-1)

6.R.P.A.3

Use ratio and rate reasoning to solve real-world and mathematical problems. (MS-PS1-1)

8.EE.A.3

Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other. (MS-PS1-1)

Viewing Options

- Hide Popup
- Black and White
- Practices and Core Ideas
- Practices and Crosscutting Concepts

by the end of the grade the SWBAT statements:

Related Evidence Statements

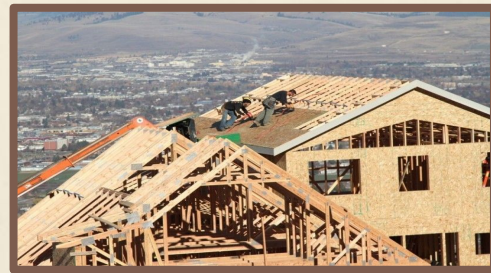
MS-PS1-1 Evidence Statements

Related Quality Units

Middle School: OpenSciEd Unit 7.3: How Do Things Inside Our Bodies Work Together to Make Us Feel the Way We Do?

CCSS

What is a Performance Expectation (PE)?

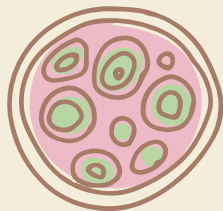


- ❑ Statements of what students should know and be able to do AFTER (6-8) and (9-11). Statements build on each other through the grade levels.
- ❑ A statement that combines practices, core ideas, and crosscutting concepts into a single statement describing how students can show what they have learned
- ❑ An NGSS is a collection of several performance expectations describing what students should be able to do to master the standard



Related Evidence Statements

MS-PS1-1 Evidence Statements



MS-PS1-1 Matter and its Interactions

Students who demonstrate understanding can:

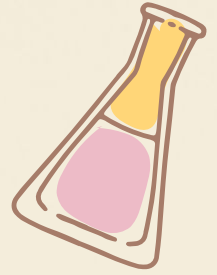
MS-PS1-1. Develop models to describe the atomic composition of simple molecules and extended structures. [Clarification Statement: Emphasis is on developing models of molecules that

Observable features of the student performance by the end of the course:

1	Components of the model
a	Students develop models of atomic composition of simple molecules and extended structures that vary in complexity. In the models, students identify the relevant components, including: - Individual atoms. - Molecules. - Extended structures with repeating subunits. - Substances (e.g., solids, liquids, and gases at the macro level).
2	Relationships
a	In the model, students describe* relationships between components, including: - Individual atoms, from two to thousands, combine to form molecules, which can be made up of the same type or different types of atom. - Some molecules can connect to each other. - In some molecules, the same atoms of different elements repeat; in other molecules, the same atom of a single element repeats.
3	Connections
a	Students use models to describe* that: - Pure substances are made up of a bulk quantity of individual atoms or molecules. Each pure substance is made up of one of the following: - Individual atoms of the same type that are connected to form extended structures. - Individual atoms of different types that repeat to form extended structures (e.g., sodium chloride). - Individual atoms that are not attracted to each other (e.g., helium). - Molecules of different types of atoms that are not attracted to each other (e.g., carbon dioxide). - Molecules of different types of atoms that are attracted to each other to form extended structures (e.g., sugar, nylon). - Molecules of the same type of atom that are not attracted to each other (e.g., oxygen). - Students use the models to describe* how the behavior of bulk substances depends on their structures at atomic and molecular levels, which are too small to see.



Where Can I Find Evidence Statements



Middle School Evidence Statements

High School Evidence Statements

*If you have *not* joined this Canvas Course before, please do so first by adding yourself to the course with the relevant link below:

- [Join TVUSD Middle School Science](#)
- [Join TVUSD High School Science](#)



Middle School CAST 2022



BVMS

39%



DMS

47%



GMS

46%



MMS

25%



TMS

43%



VRMS

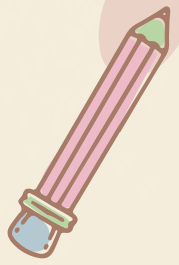
43%



HIIA MS

35%

High School CAST 2022



GOHS

46%

RVHS

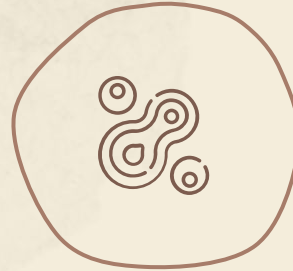
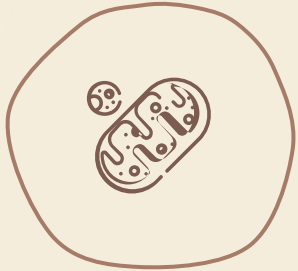
11%

CHS

38%

TVHS

39%

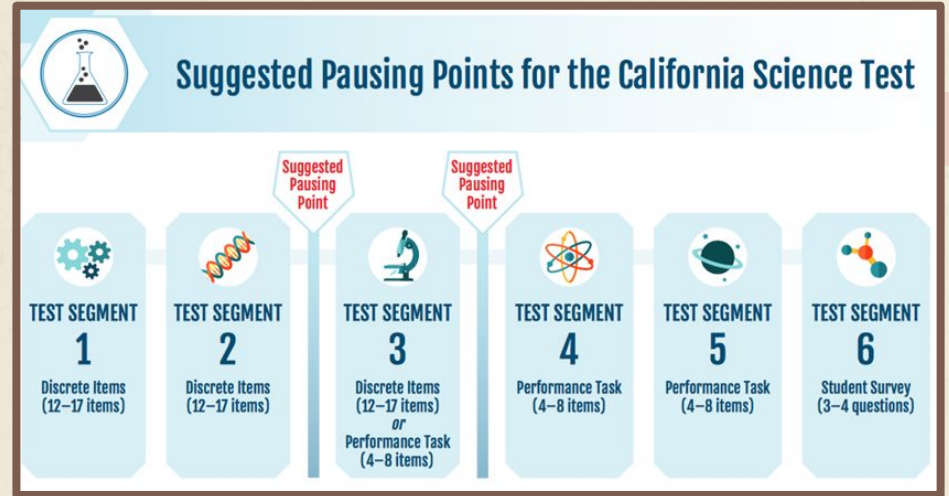


CAST Assessment

About CAST

TVUSD Middle School Science

TVUSD High School Science



How Does NGSS Look in TVUSD?



Semester 1	Integrated Learning Segment and Phenomenon	ID	Unit & Lessons *From TCI unless otherwise noted	Lesson Phenomenon as a question	PEs	DCIs	SEPs	CCCs	Common Summative & Performance Assessments	Non-TCI Resources
	These lessons are NOT in TCI									
	Segment 0: Engage in the Nature of Science How are some students more successful in science than others?	6S0U0	L0.1: Canvas and IC							Canvas Overview How To Use Canvas for Students How to Become a Parent Observer
			L0.2: SEPs							SEP Slides SEP Foldable
			L0.3: CCCs							CCC Slides CCCs Foldable
			(L0.4: Lab Safety)							Sample Lab Safety Contract
			(L0.5 Measurement)							Flinn's Contract
	Segment 1: Systems & Subsystems in Earth & Life Science Sometimes, people get sunburned skin from a day at the beach. How does this happen?	6S1U1 Atmosphere	Unit 1: The Atmosphere & Energy							
			L1-Earth's Atmosphere							
			L2-Taking Earth's Temp.							
		6S1U3 Weather	Engineering Challenge: Heat Transfers cooler/solar oven							
			L3-Earth & Solar Energy							
			L6- Air Pressure and Wind							
		6S1U3 Weather	L7- Water and the Weather							
			L8- Air Masses and Changing Weather							
			L9-Severe Weather							



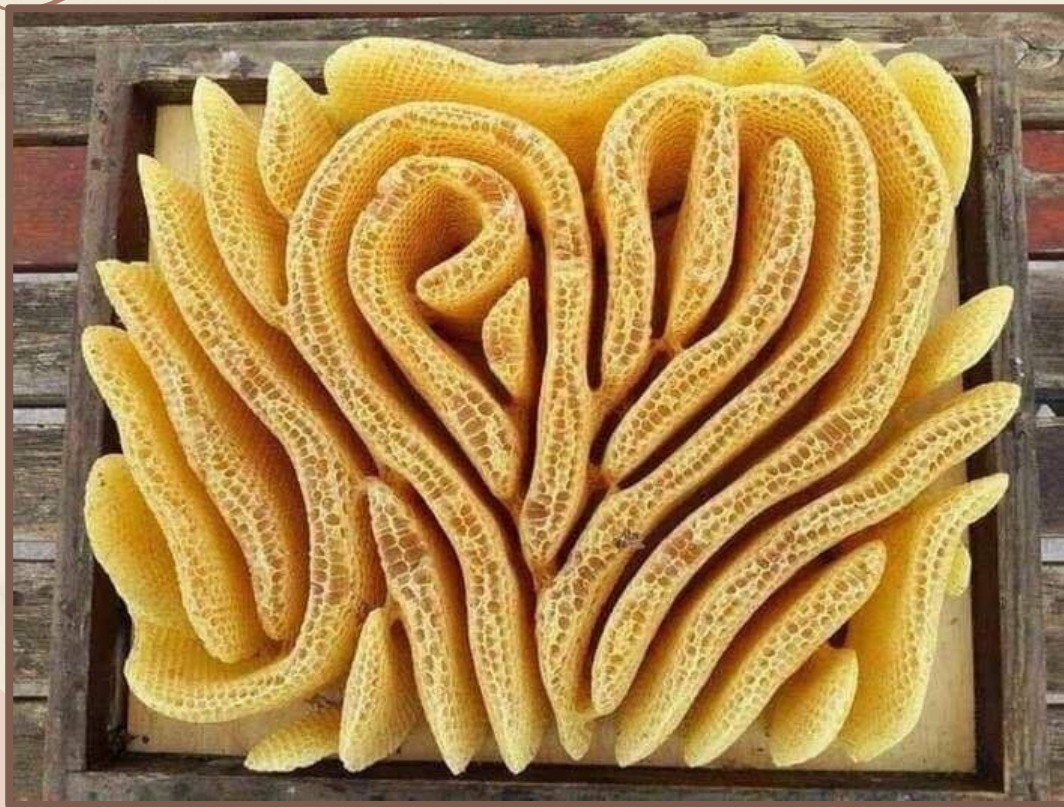
SCIENCE EDUCATION WILL INVOLVE LESS:	SCIENCE EDUCATION WILL INVOLVE MORE:
Rote memorization of facts and terminology	Facts and terminology learned as needed while developing explanations and designing solutions supported by evidence-based arguments and reasoning.
Learning of ideas disconnected from questions about phenomena	Systems thinking and modeling to explain phenomena and to give a context for the ideas to be learned
Teachers providing information to the whole class	Students conducting investigations, solving problems, and engaging in discussions with teachers' guidance
Teachers posing questions with only one right answer	Students discussing open-ended questions that focus on the strength of the evidence used to generate claims
Students reading textbooks and answering questions at the end of the chapter	Students reading multiple sources, including science-related magazine and journal articles and web-based resources; students developing summaries of information.
Pre-planned outcome for “cookbook” laboratories or hands-on activities	Multiple investigations driven by students' questions with a range of possible outcomes that collectively lead to a deep understanding of established core scientific ideas
Worksheets	Student writing of journals, reports, posters, and media presentations that explain and argue
Oversimplification of activities for students who are perceived to be less able to do science and engineering	Provision of supports so that all students can engage in sophisticated science and engineering practices

What Should You See in a Lesson?

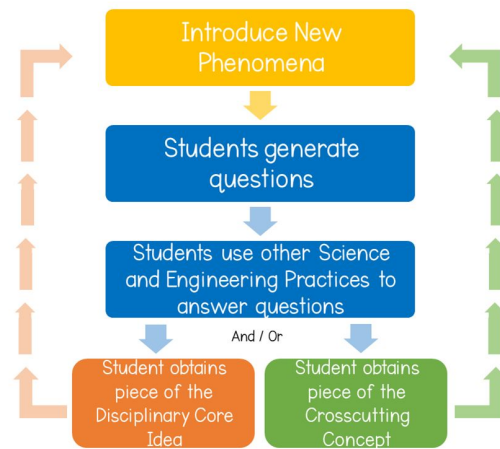
- ★ Student Centered
- ★ Student Movement and Exploration
- ★ Various correct answers from students
- ★ Student Collaboration
- ★ Inquiry driven by Phenomena



What about PHENOMENA?



USE PHENOMENA TO DRIVE 3-DIMENSIONAL INSTRUCTION



and student experiences will
drive understanding!



Engage

This lesson mentally engages students with an activity or question. It captures their interest, provides an opportunity for them to express what they know about the concept or skill being developed, and helps them to make connections between what they know and the new ideas.

Explore

Students carry out hands-on activities in which they can explore the concept or skill. They grapple with the problem or phenomenon and describe it in their own words. This phase allows students to acquire a common set of experiences that they can use to help each other make sense of the new concept or skill.

Explain

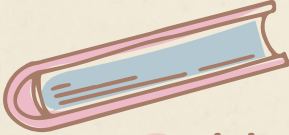
Only after students have explored the concept or skill does the teacher provide the concepts and terms used by the students to develop explanations for the phenomenon they have experienced. The significant aspect of this phase is that explanation follows experience.

Elaborate

This phase provides opportunities for students to apply what they have learned to new situations and so develop a deeper understanding of the concept or greater use of the skill. It is important for students to discuss and compare their ideas with each other during this phase.

Evaluate

The final phase provides an opportunity for students to review and reflect on their own learning and new understandings and skills. It is also when students provide evidence for changes to their understandings, beliefs and skills.



How Can You Support NGSS Implementation?

Canvas

- Redirect teachers to Canvas for all needed materials

HS Pilot

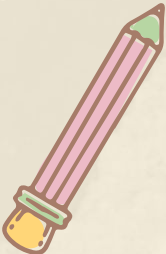
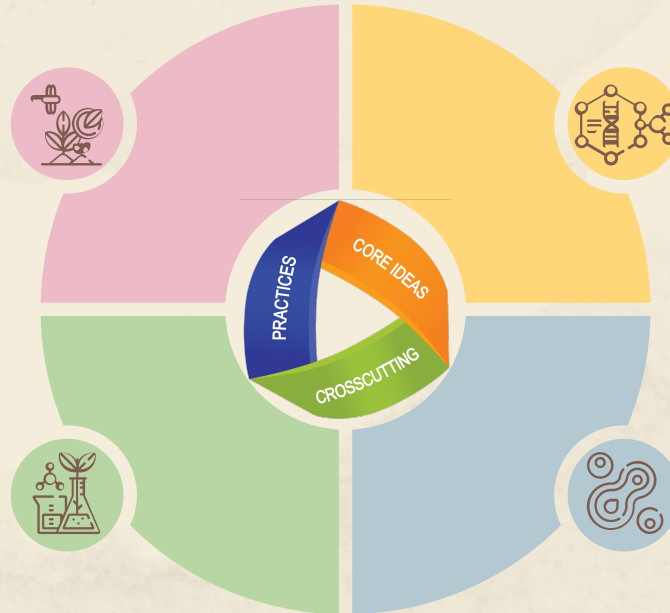
- HMH through Dec.
- Decision 1/19/22

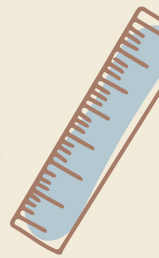
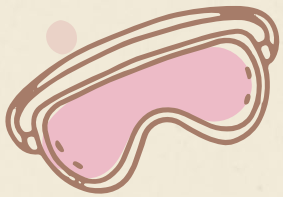
Site

- Feedback with walkthroughs
- PLCs around the Evidence Statements
- Develop your understanding of NGSS

MS Design Team

- November, 2 more days
- Integrated Science



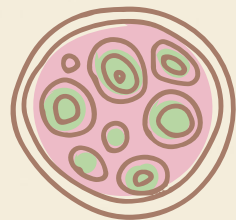


Strand Release Day

OTHER

Safety Release

Risk Management



How Will You Support Changes in Teaching and Thinking Differently Around Science?

